Dr. Alex Skeels

Week 8: Species in Space II

More Species Distribution Modelling

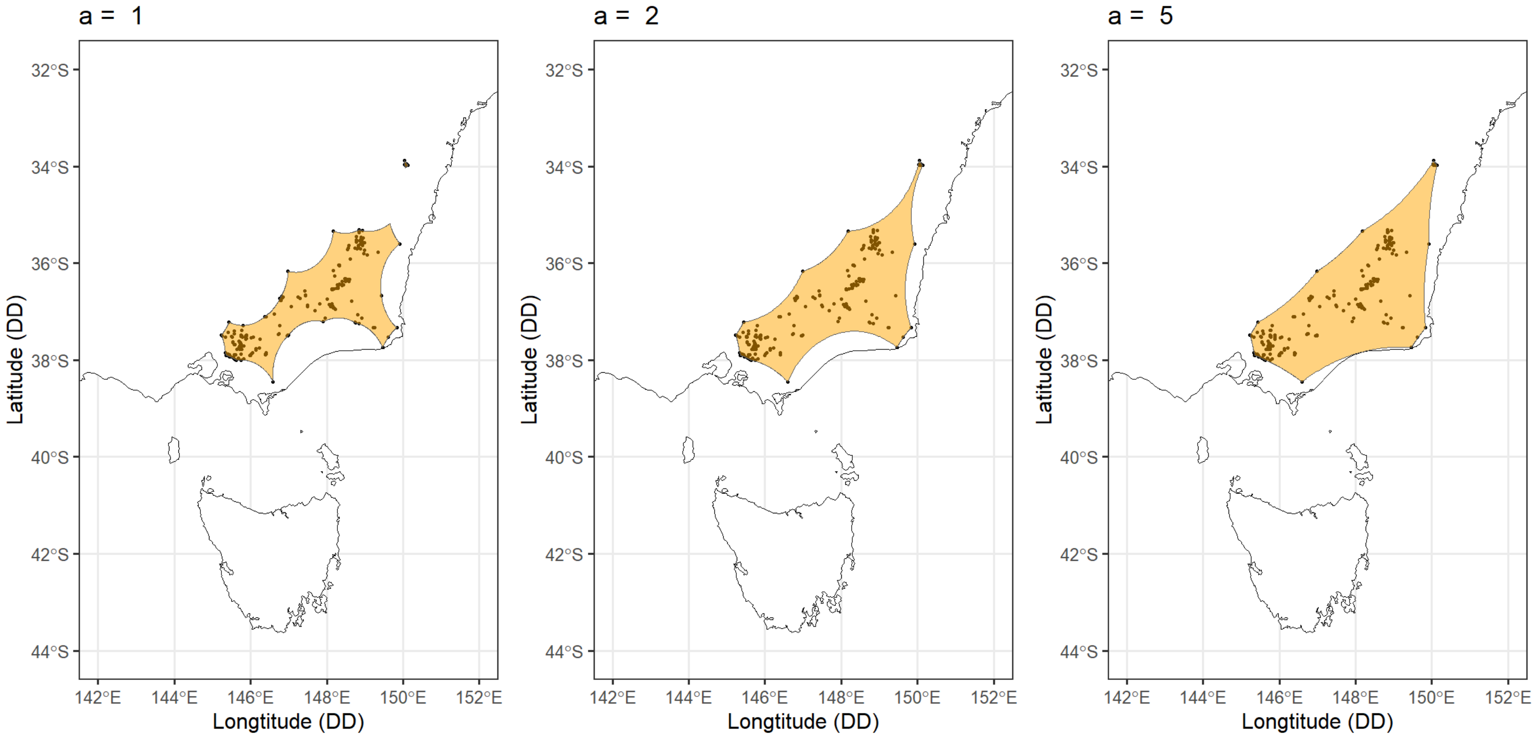
BIOL3207 – Data Science for Biologists

Weeks 7 and 8

Date	Session	Subject
Week 7	Lecture 1	Species in Space I: An introduction to species distribution modelling
Week 7	Workshop 1	Points, polygons, and geometric SDMs in R
Week 8	Lecture 2	Species in Space II: More species distribution modelling
Week 8	Workshop 2	Climate data and statistical SDMs in R

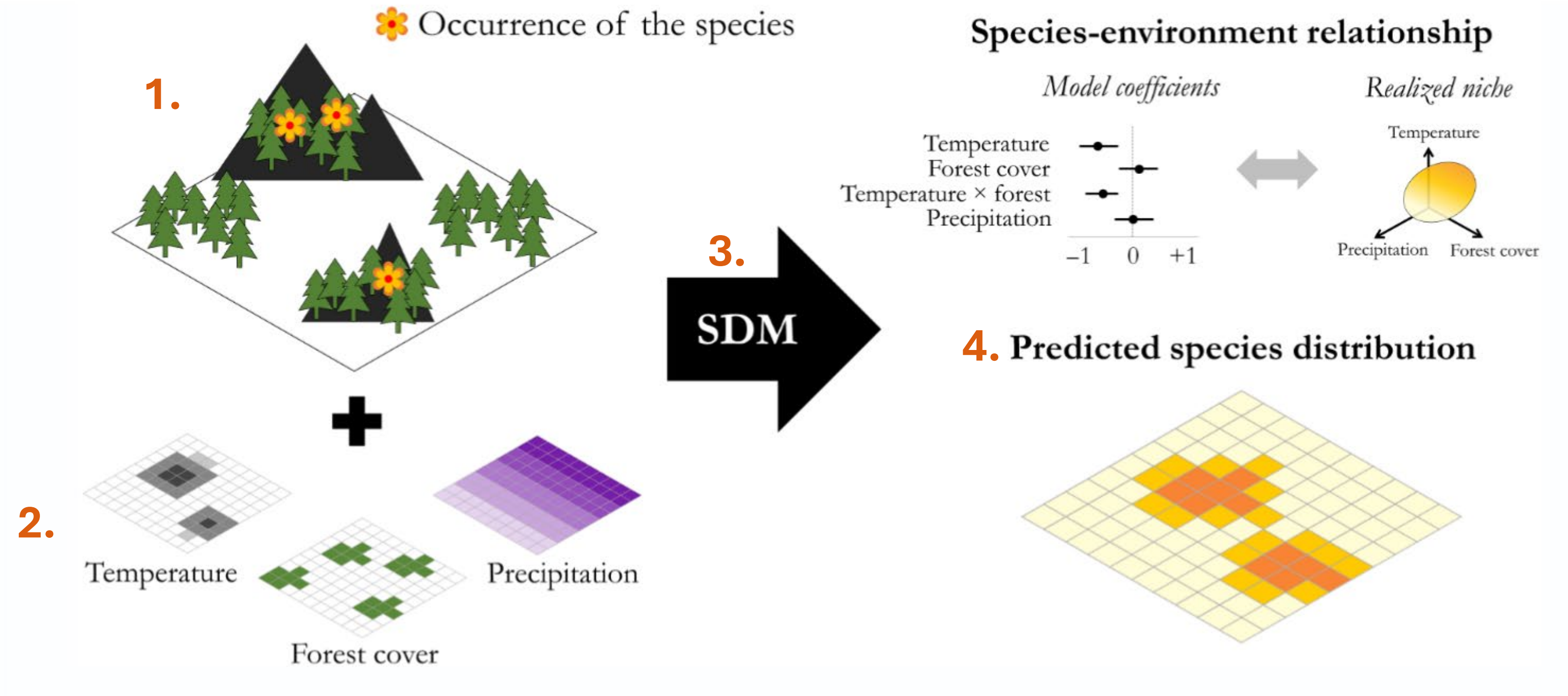
Species Distribution Models

Last week: Geometric SDMs (hulls and buffers) as model-free methods to estimate ranges.



Species Distribution Models

This week: Statistical SDMs model the association of presence and absence records with environmental variables



Riva et al. 2024. Journal of Ecology

1. Presence-absence data

2. Gridded environmental data

3. Statistical model

4. Prediction on new data

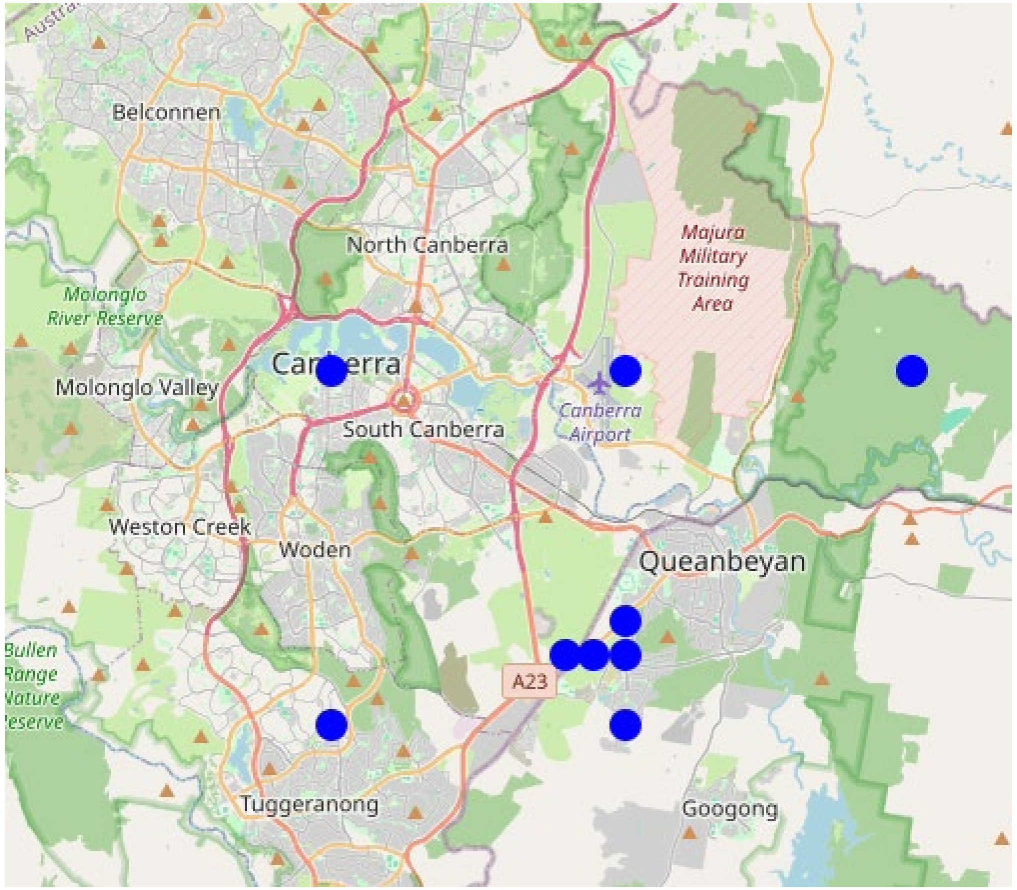
Data Preparation

Presence/absence data, environmental predictors, and variable selection

Presence-Absence Data

- Presence points from direct observations (Week 11)
- Where does absence data come from?
    - Field surveys
- Problems:
    - Easier to survey absences for conspicuous species that don't move, e.g. **trees**
    - Harder for others, e.g., cryptic, ephemeral, vagile, and small organisms
- *Just because you don't see something doesn't mean it's not there*

Where *isn't* this species distributed?



Canberra Grassland Earless Dragon
Tympanocryptis lineata

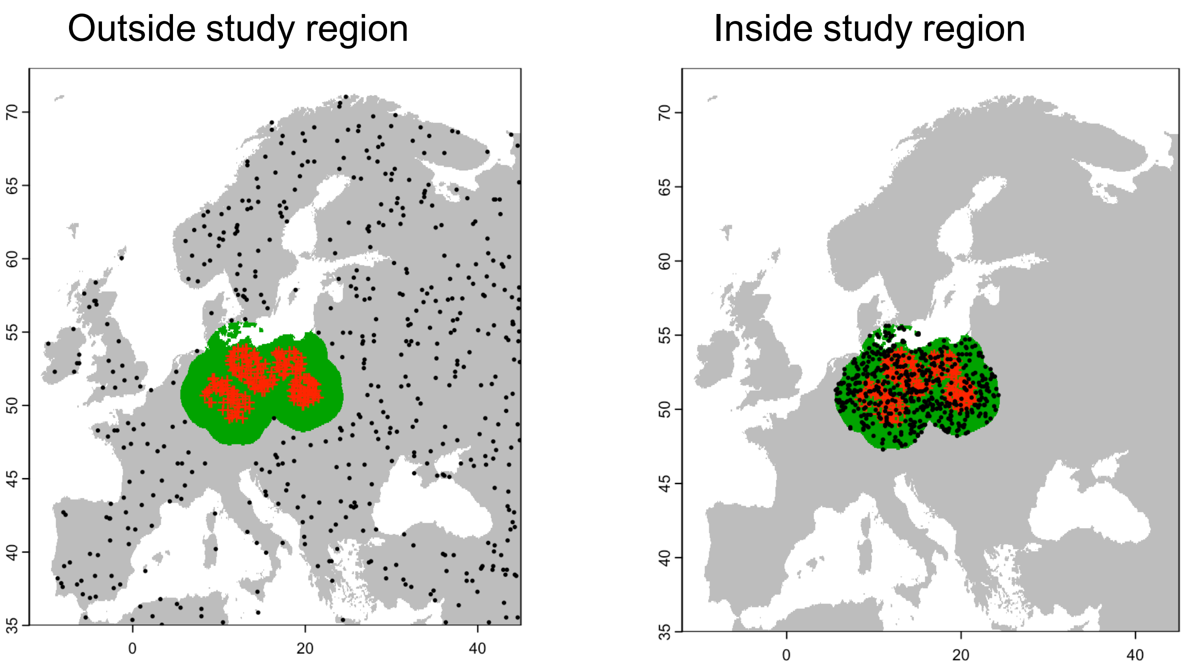
Absence Points

- One solution: pseudo-absence or "background" data
- Pseudo-absence are points sampled in areas with likely true absences
- There is no "best way" to sample pseudo-absences and the number of points + where they are sampled from depends on the method you are using
- Testing the sensitivity of SDM to choice of pseudo-absence sampling methods is common practice

Methods in Ecology and Evolution
British Ecological Society
Methods in Ecology and Evolution 2012, **3**, 327–338 doi: 10.1111/j.2041-210X.2011.00172.x

Selecting pseudo-absences for species distribution models: how, where and how many?

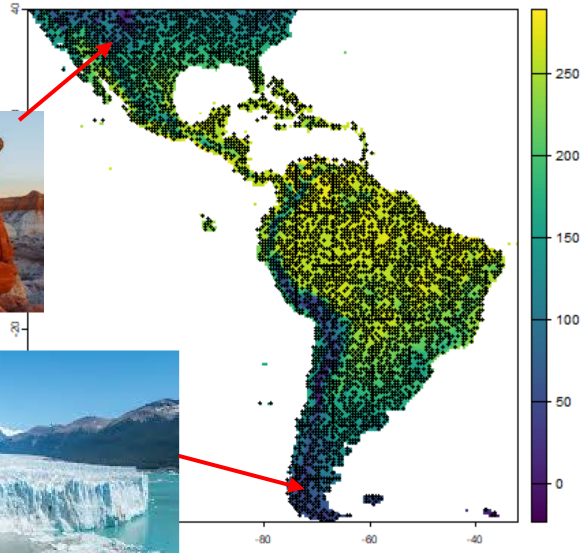
Morgane Barbet-Massin[1]*, Frédéric Jiguet[1], Cécile Hélène Albert[2,3] and Wilfried Thuiller[3]



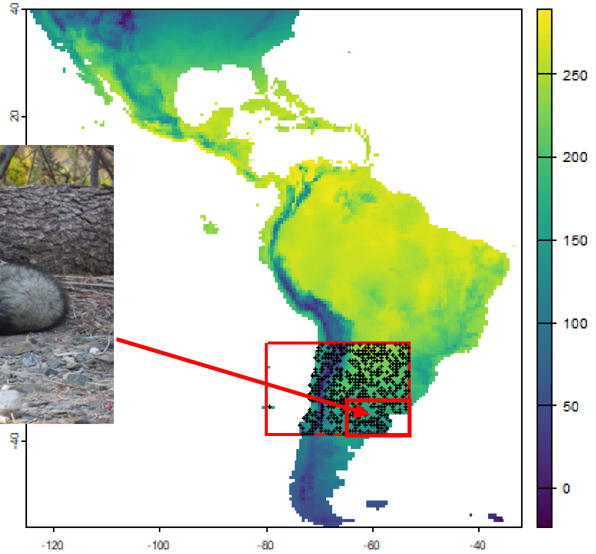
https://damariszurell.github.io/EEC-MGC/b5_pseudoabsence.html

Absence Points

- **The "*background*" defines the question you're asking**
- An SDM learns what separates presences from absences. Change the background, change the answer.
- **Too restrictive** and presences and background overlap in environmental space which leads to weak, overfit model
- **Too broad** and we risk trivial contrasts, and poor real-world transfer
- **Ideal** → somewhere the species plausibly *could* occur but hasn't been recorded

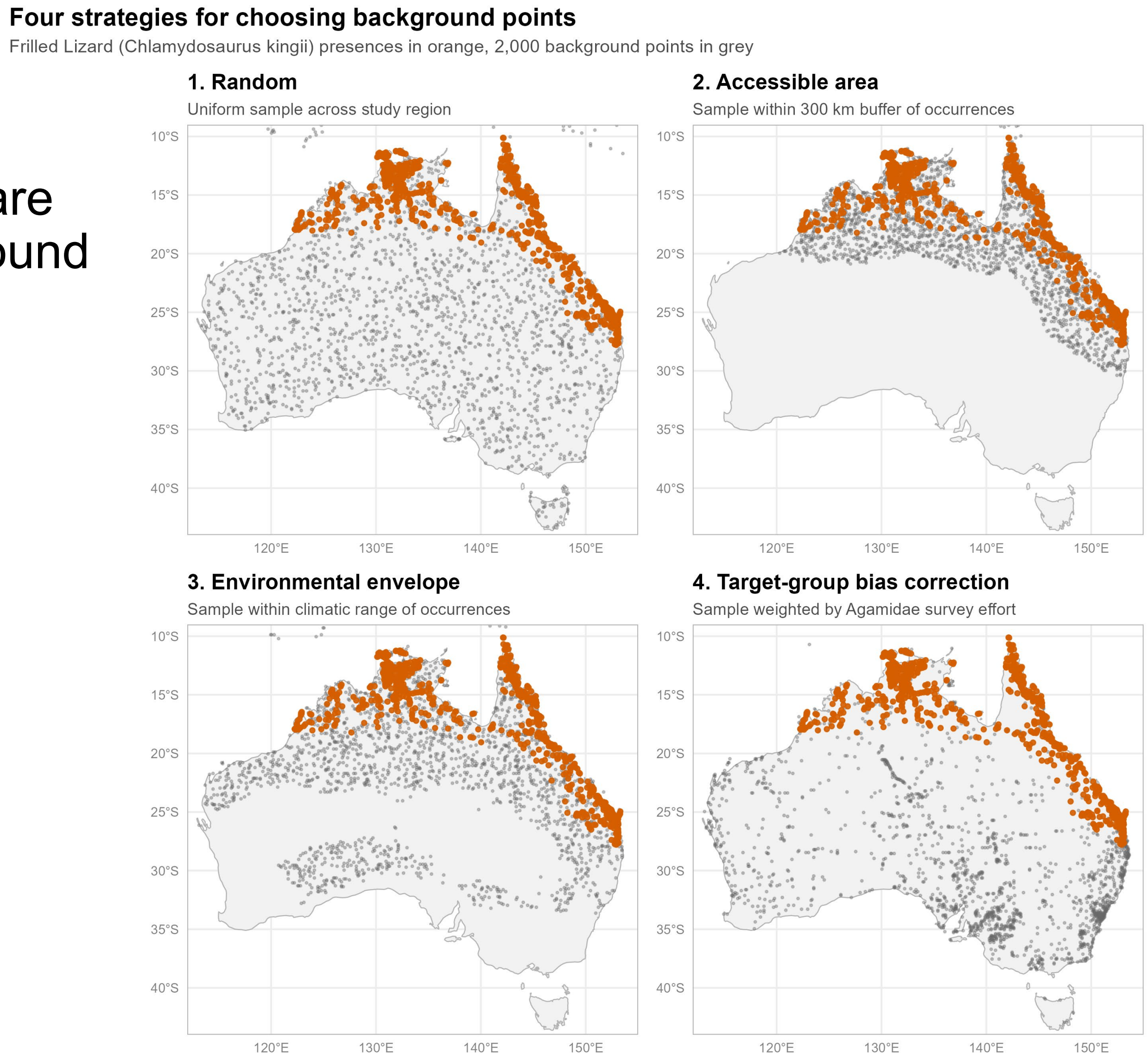


Pampas Fox



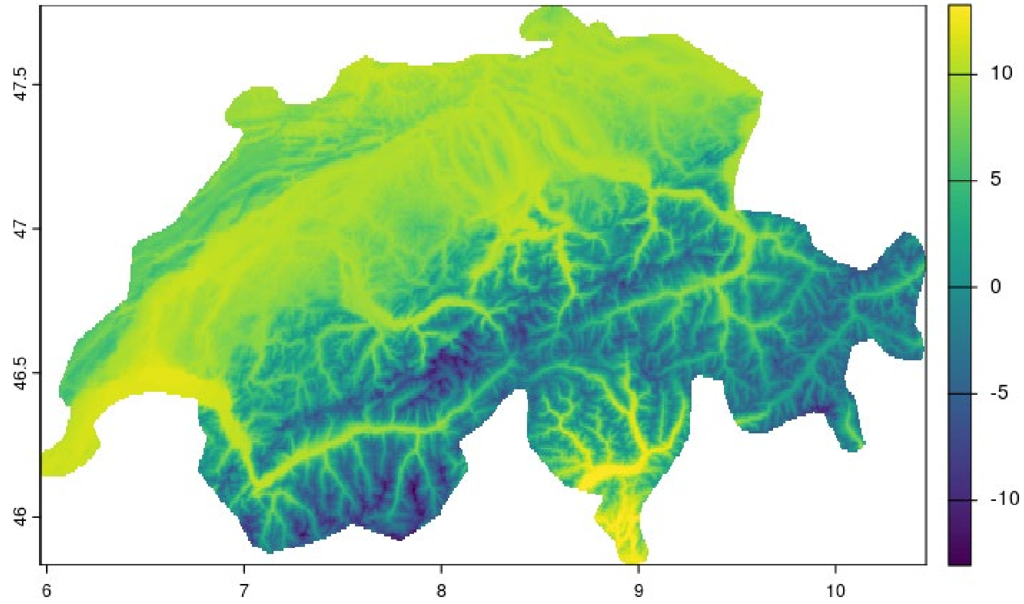
Absence Points

- In this week's workshop we compare four methods for selecting background points for our Frilled Lizards!



Predictor Variables: Rasters

- SDMs use environmental data as predictors of suitable habitat
- Environmental data is often in a gridded format known as a **raster**
- Each pixel is associated with a value which can be plotted with colour to create maps
- Global–regional scale environmental data available at high resolution
  - CHELSA, WorldClim, NASA , FAO



Mean Annual Temperature: Brun et al. 2022

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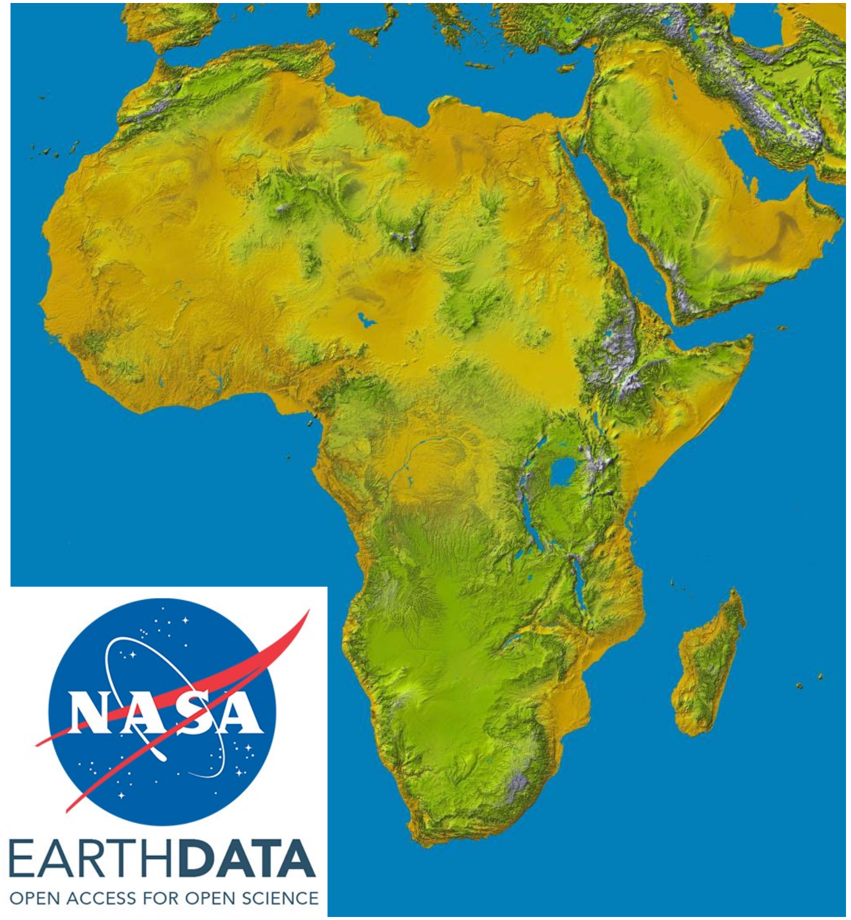
Shuttle Radar Topography Mission, NASA



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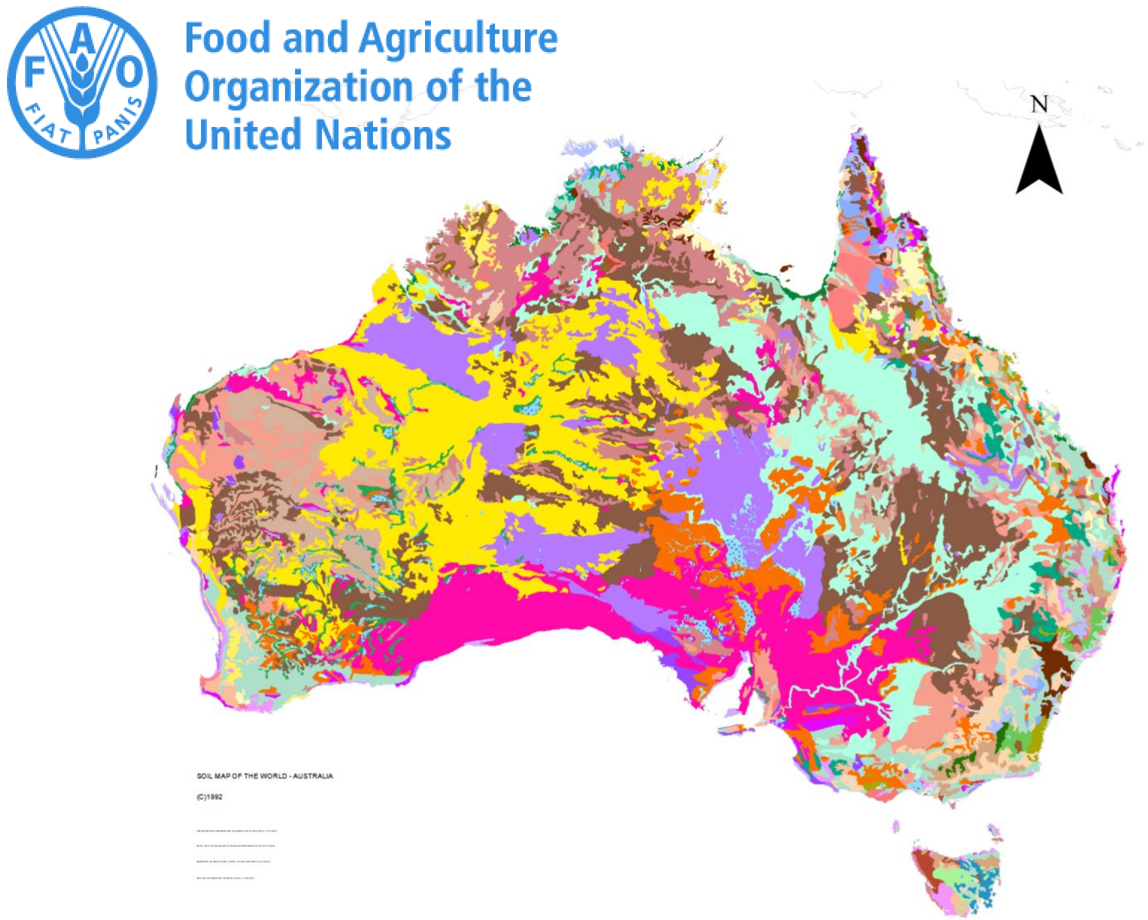
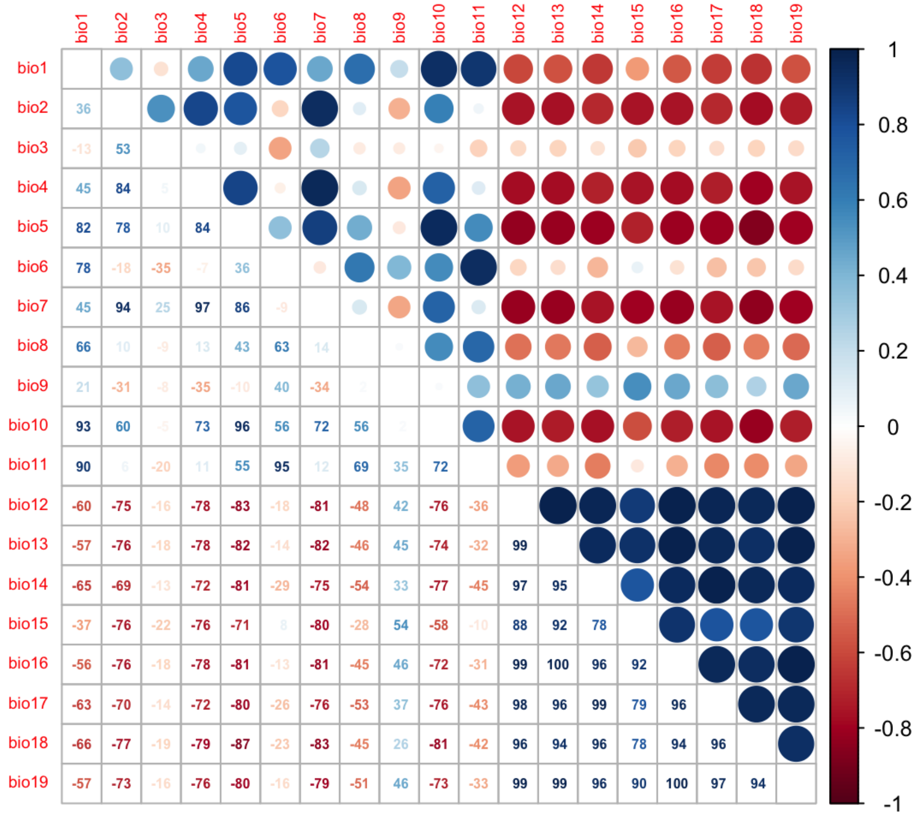
Soil Map of the World



Predictor Variables: Selection

- **Choosing predictors matters**
- Many variables to choose from
- 19 bioclim variables (common temperature and precipitation measures)
  - many are highly correlated (mean/min/max temp)
- Correlation breaks coefficient estimates and inflates variable importance
- **Two approaches, use both:**
  - **Ecological.** pick variables that plausibly limit the species
  - **Statistical.** drop redundant predictors (|r| > 0.7), compare model fit

Correlation Matrix of 19 BioClim Variables



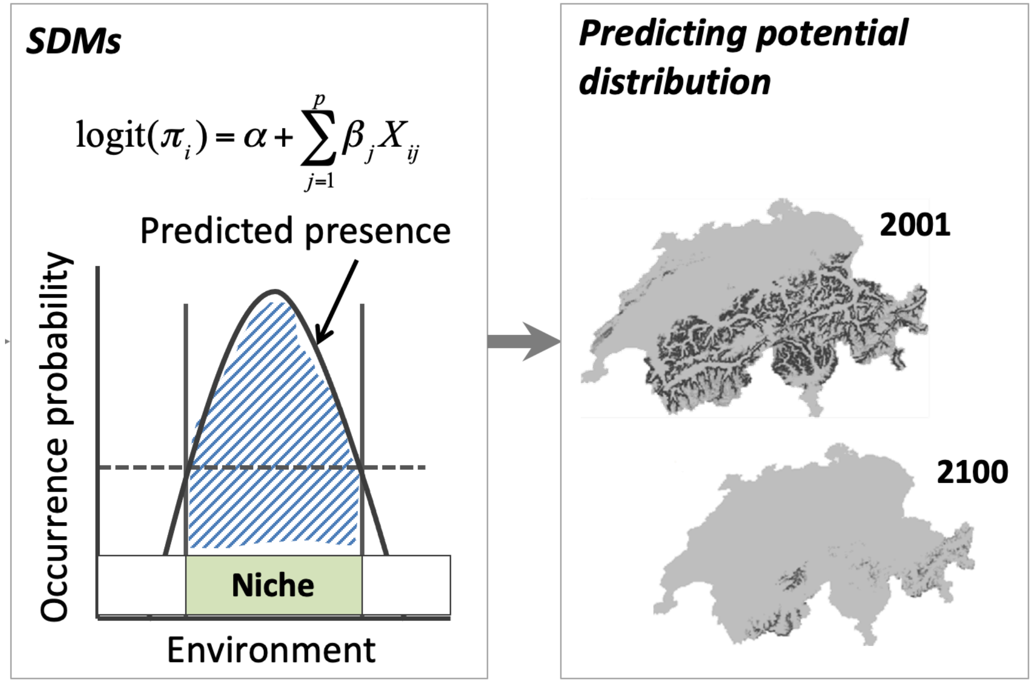
Zurell, 2023

Model Fitting

GLMs, GAMs, Maxent, and machine learning algorithms

Statistical Models

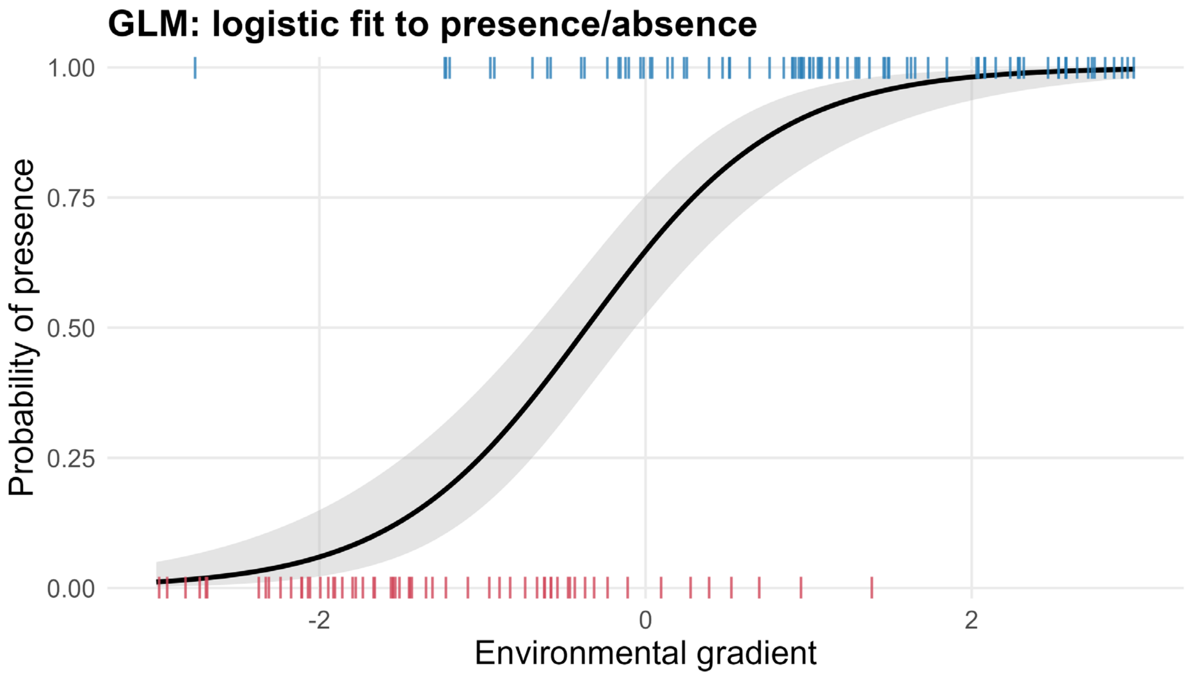
- **The question:** *"Given the climate here, how likely is this species to be found?"*
- An SDM learns that relationship from where the species **has** been seen (presences) vs where it plausibly **could have** been but wasn't (background points).
- The model gives every cell on the map a **suitability score from 0 to 1**.
- Different algorithms draw this relationship differently — today we'll use two:
  - **GLM** — assumes simple straight-line effects
  - **GAM** — allows curves



Zurell, 2023

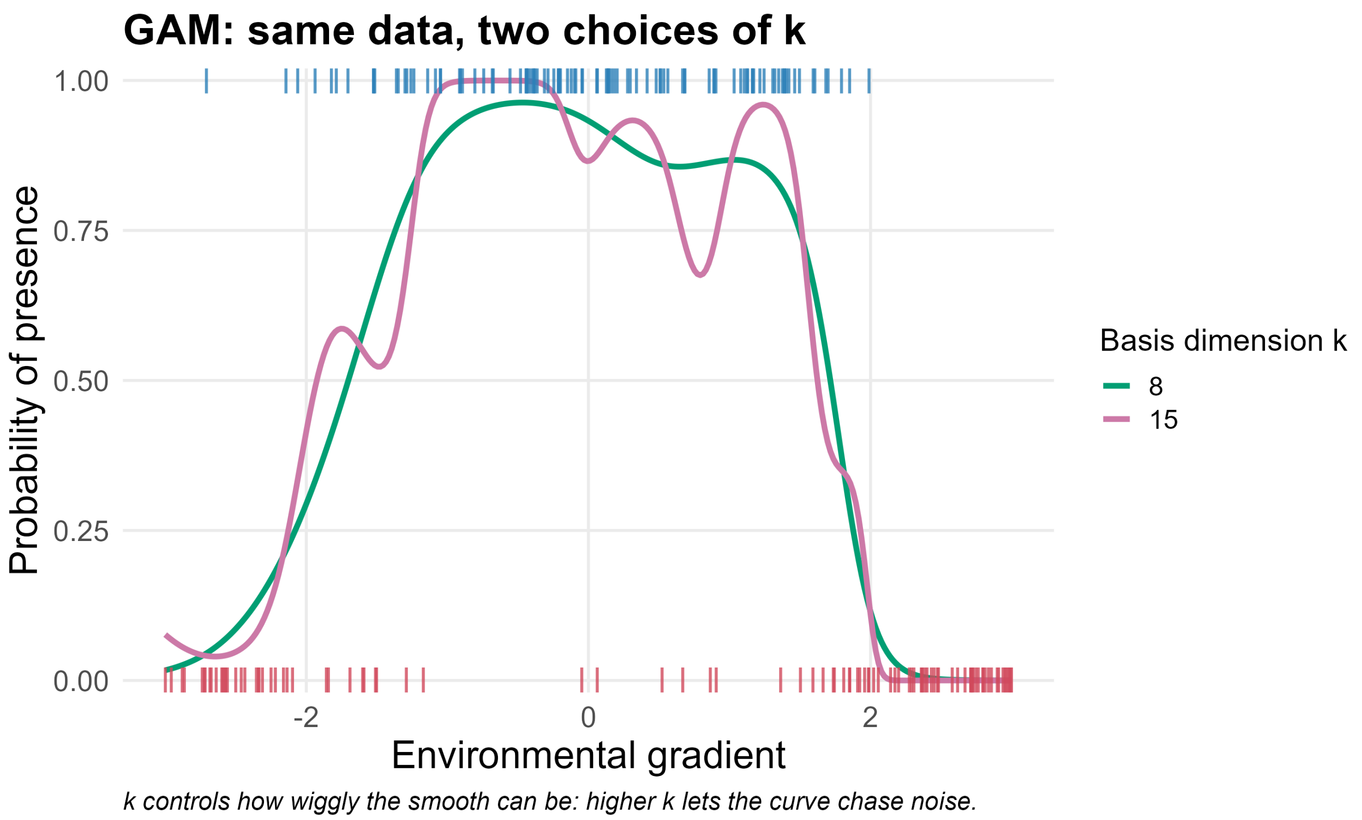
GLM: straight-line models

- **Idea:** each environmental variable has a **steady effect** —
- The GLM fits an **S-shaped curve** (so predictions stay between 0 and 1) where the slope is controlled by each variable.
- Reading the output:
  - A **positive coefficient** → higher values of that variable = more suitable
  - A **negative coefficient** → higher values = less suitable
  - A **big coefficient** → that variable matters a lot
- **Strength:** easy to interpret — you can say *"warmer sites are more suitable“*
- **Weakness:** can't handle "likes the middle, hates the extremes"



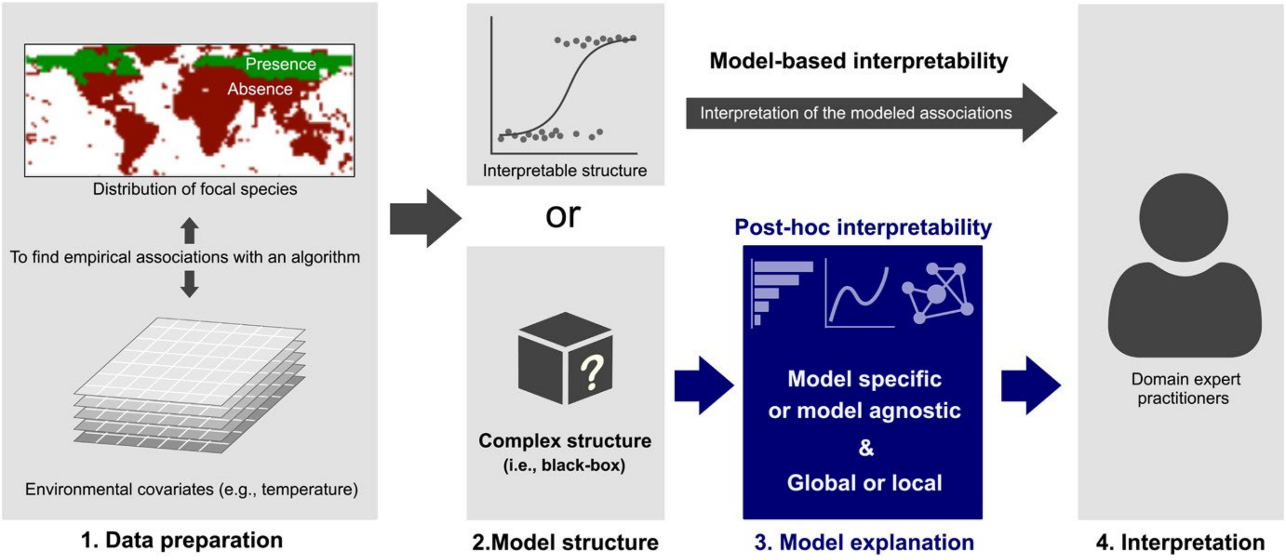
GAM: bendy-line models

- A straight line can't capture *"good in the middle, bad at both ends"*.
- **GAM** lets each variable bend — it fits a **flexible curve** instead of a line.
- The knob k controls how bendy: k = 2 is almost straight, k = 10 is very wiggly
- Too wiggly = the model starts chasing noise (overfitting)
- **Strength:** captures realistic unimodal responses (optima, tolerances)
- **Weakness:** harder to summarise in a single sentence; risk of overfitting



Machine Learning: black-box models

- A range of machine-learning methods exist for SDMs — Random Forests, Boosted Regression Trees, MaxEnt, neural networks.
- They can capture **highly complex, non-linear relationships** and interactions between predictors that GLMs and GAMs cannot.
- Trade-off: these models are **harder to interpret** — less transparent about *why* a prediction was made ("black-box" models).



"It is reasonable to ask whether ecologists should sacrifice interpretability by using excessively complex algorithms for constructing SDMs in order to procure slight advantages in predictive accuracy" *Ryo, et al. 2020 Ecography*

Model Predictions

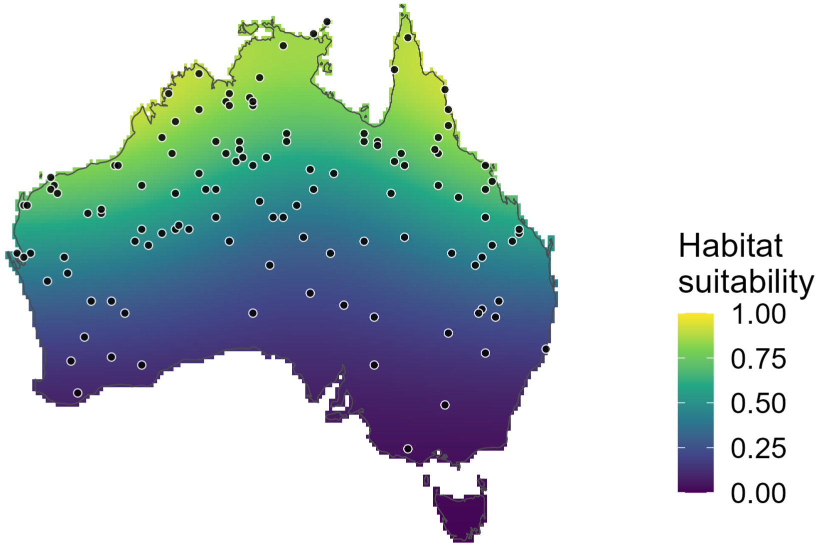
Range maps, thresholding, forecasting, hindcasting

Suitability

- The output of an SDM is a **continuous map** — every pixel gets a score between 0 (unsuitable) and 1 (highly suitable).
- The score measures how **similar a location's environment is** to the places the species has been recorded.
- Because we used pseudo-absences, this is **relative habitat suitability**, not probability of occurrence.

Predicted habitat suitability

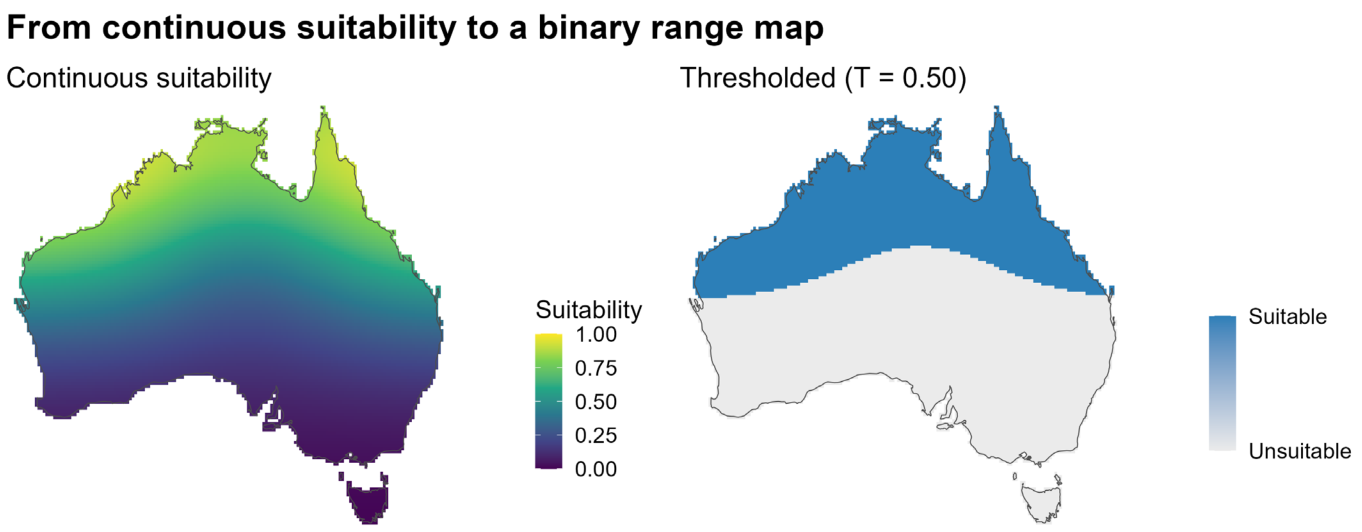
SDM output is a continuous map, 0 (unsuitable) → 1 (highly suitable)



Black dots: simulated species presence records.

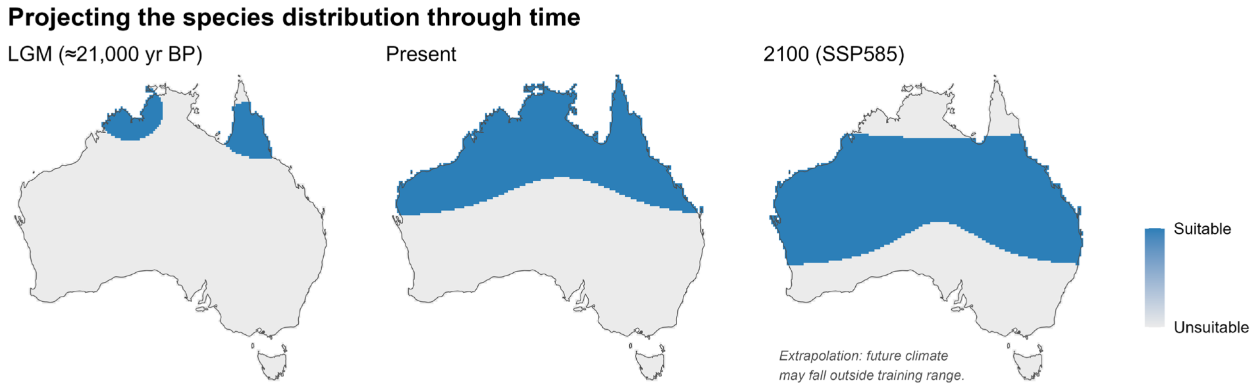
Thresholding

- Most real questions need a **yes/no answer** — is this place part of the species' range?
- Pick a **threshold** T; cells with suitability > T are "suitable", the rest "unsuitable".
- Threshold choice is a **decision, not a given**
- Low T ⇒ large predicted range (overprediction). High T ⇒ small, conservative range.



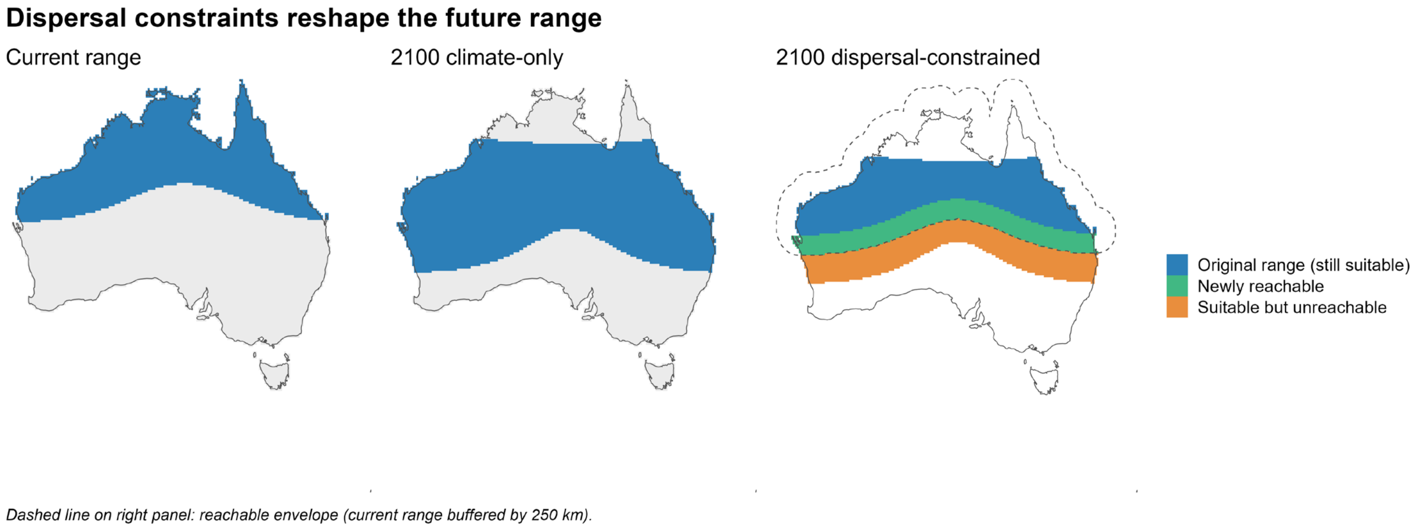
Model Projection

- The model has learned a **climate–species relationship**. Feed it climate layers from a different time and it predicts where the niche was (or will be) then.
- **LGM (~21,000 yr ago)**: contracted to refugia as conditions cooled and dried.
- **2100 under SSP585**: the niche shifts south as the tropical north becomes too hot.
- ⚠️ **Caution**: future climate may sit outside the range the model was trained on — those pixels are **extrapolations**, not interpolations.



Dispersal

- A place being **climatically suitable** in 2100 doesn't mean the species can **get there** in time.
- Real species have **limited dispersal**
- Dispersal models keep only future-suitable areas that fall within a **reachable envelope** around the current range
  - Different algorithms exist to do this
- **Climate-only projections overestimate the future range.** Dispersal models bring projections closer to reality.

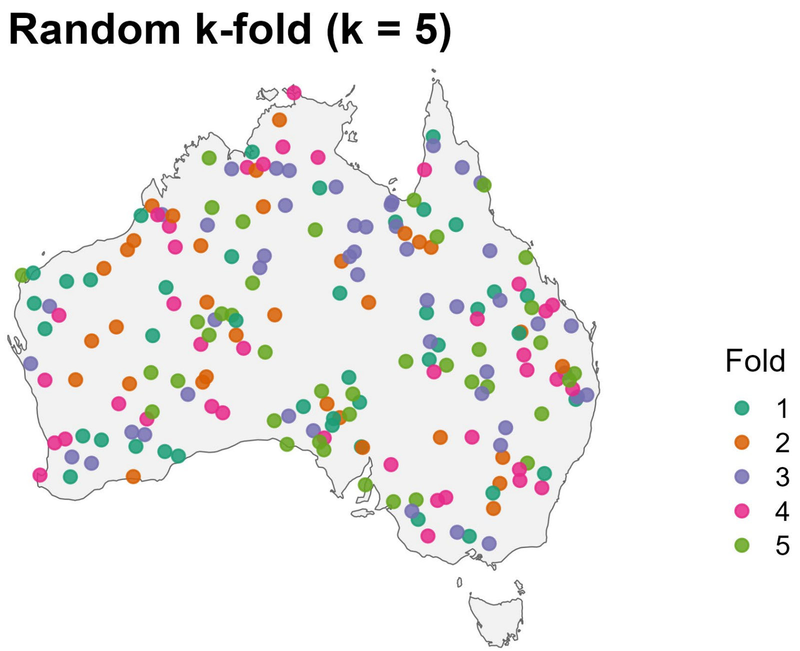


Model Assessment

Cross-validation and evaluation metrics

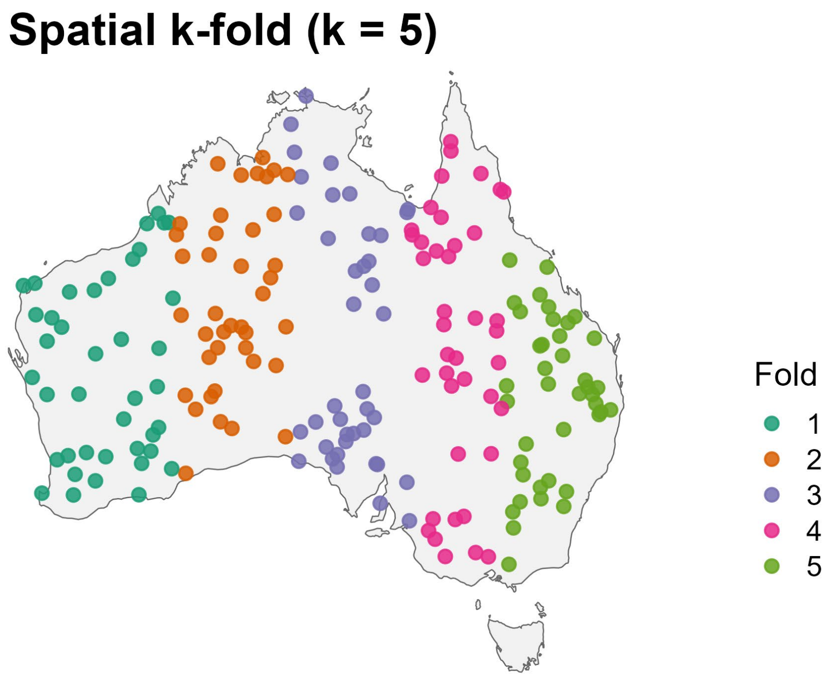
Cross-Validation

- How do we know a model has **learned** the pattern, rather than just memorised the data?
- Any model can fit its own training data perfectly — that's not evidence of anything.
- The only honest test is whether it can predict **data it has never seen**.
- We need to separate data into “training” and “testing” subsets
    - *k*-fold cross validation repeated n times
- **Overfit models look great on training data but fail on new data.**



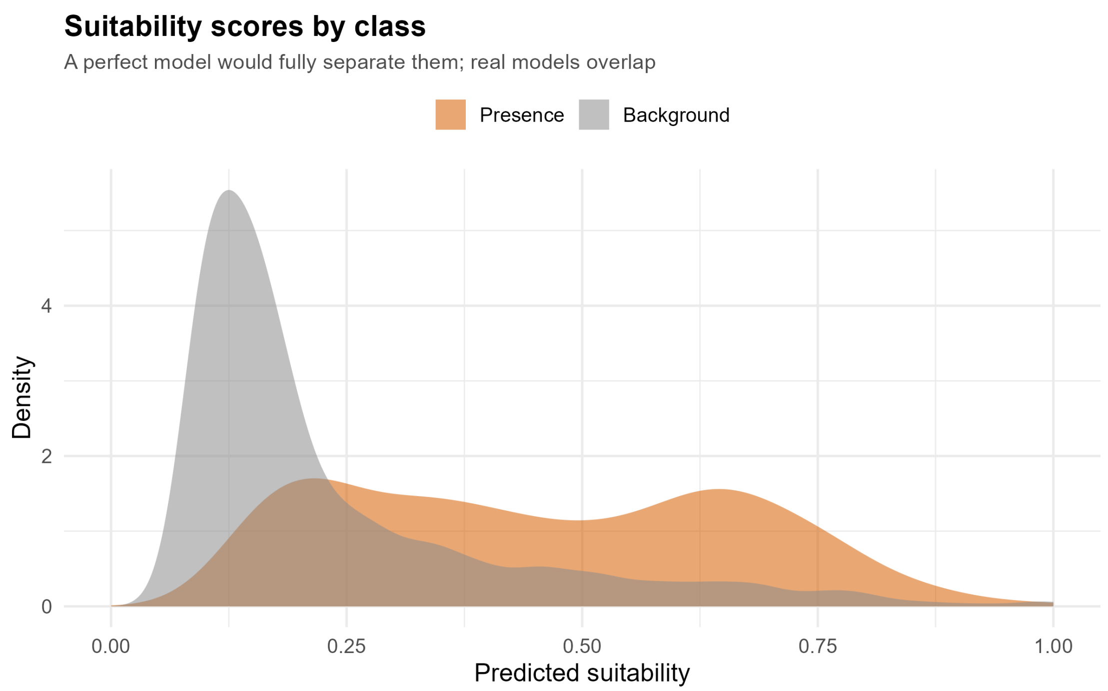
Cross-Validation

- How do we know a model has learned the species' **niche**?
- Many models can predict well for points near its training data.
- The only honest test is whether it can predict **a region it has never seen**.
- Random k-fold can look great because nearby points are easy to predict — *spatial k-fold doesn't let the model cheat*.



Model Evaluation

- Does the model give *higher* scores at places the species has actually been seen than at places it hasn't?
- If the two curves **fully separated**, the model would be perfect
- If they **fully overlapped**, the model would be useless
- Real models sit somewhere in between — and the overlap is where evaluation matters
- *Evaluation = measuring how well those two distributions separate.*



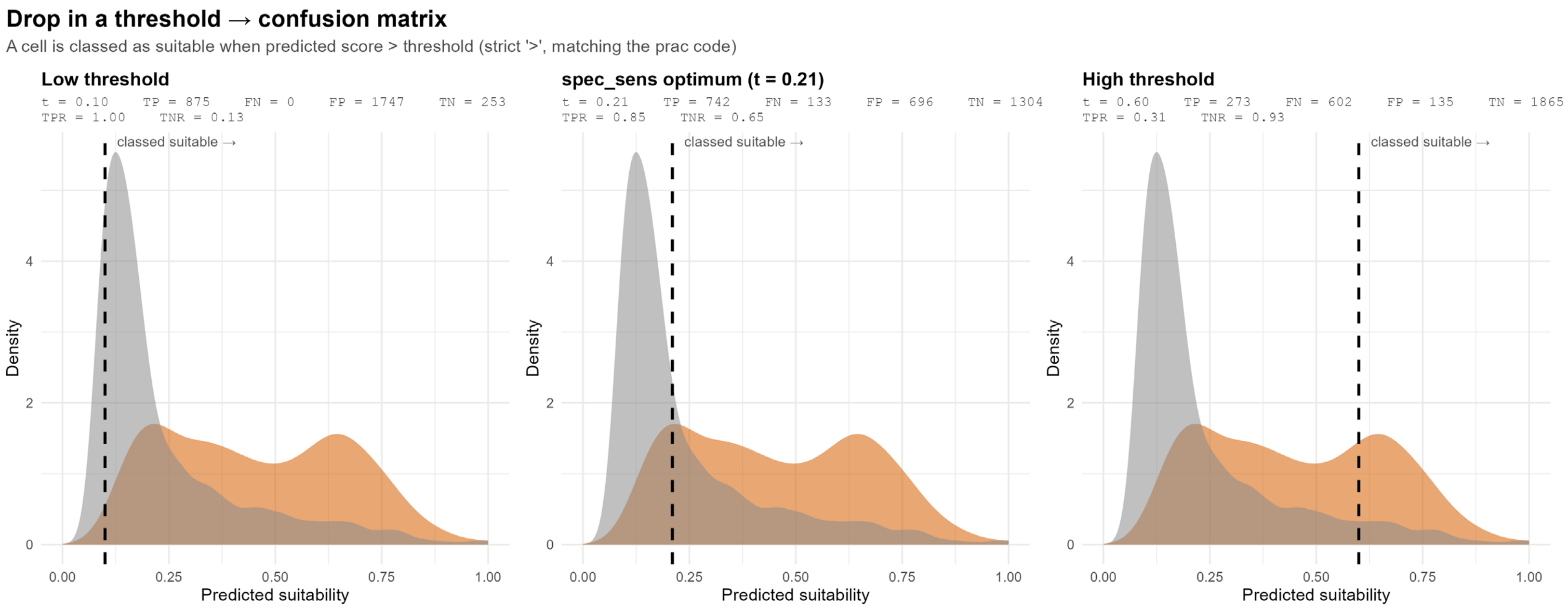
Model Evaluation

- **Confusion matrix**
- Turn predictions into four counts. At any chosen threshold:
  - **TP** — species is there, model says present ✓
  - **FN** — species is there, model says absent ✗ *(model missed it)*
  - **FP** — species isn't there, model says present ✗ *(false alarm)*
  - **TN** — species isn't there, model says absent ✓
- A good model maximises the diagonal (TP + TN) and minimises the off-diagonal (FN + FP).

Observed \ Predicted	**Positive**	**Negative**
Positive	True Positive (a)	False Negative (b)
Negative	False Positive (c)	True Negative (d)

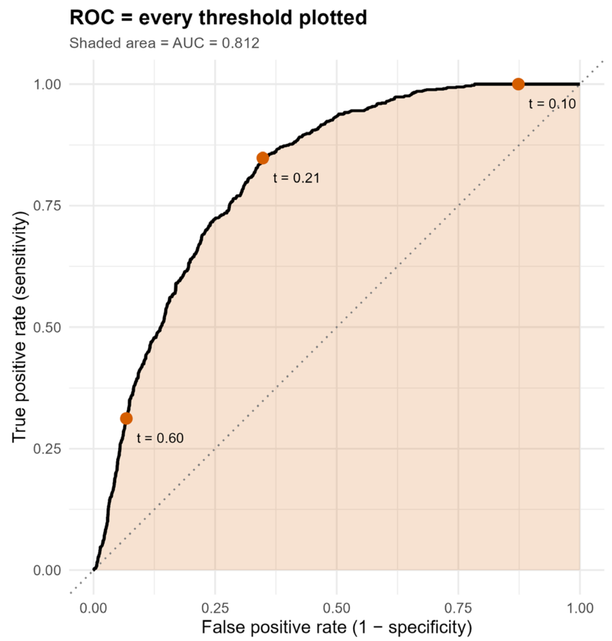
Model Evaluation

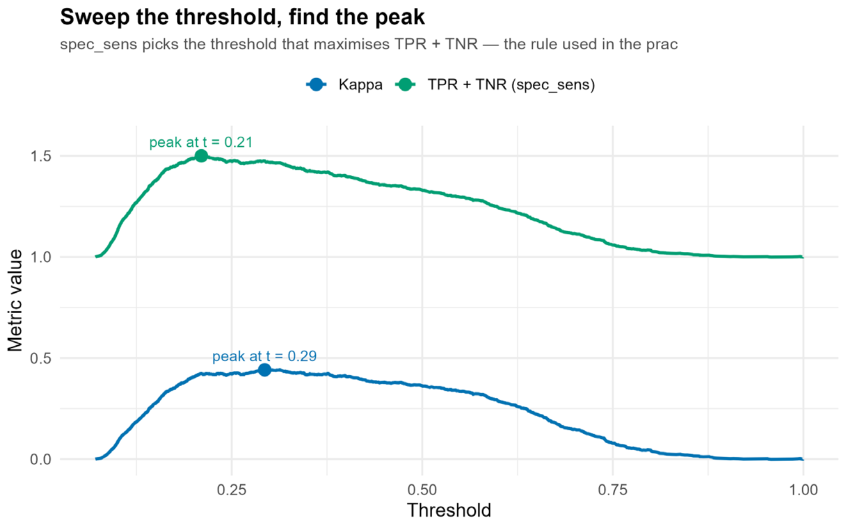
- **Different thresholds → different confusion matrices → different rates:**
- **Low threshold** — catches nearly every presence (high TPR) but flags huge areas as suitable (low TNR)
- **High threshold** — very selective (high TNR) but misses real presences (low TPR)



Model Evaluation

- Plot every possible threshold's (FPR, TPR) on one graph → the **Receiver Operator Curve (ROC)**.
- **AUC = Area Under the Curve** — one number summarising performance across *all* thresholds
  - AUC = 1 → perfect
  - AUC = 0.5 → no better than coin flip
  - AUC ≈ 0.8 → a reasonable SDM
- **But AUC alone doesn't pick a threshold.** it's an average. To draw a binary map we still need one.
- Common rules:
  - **spec_sens** — the threshold where sensitivity + specificity is highest
  - **Kappa** — peak of an accuracy metric that corrects for chance agreement

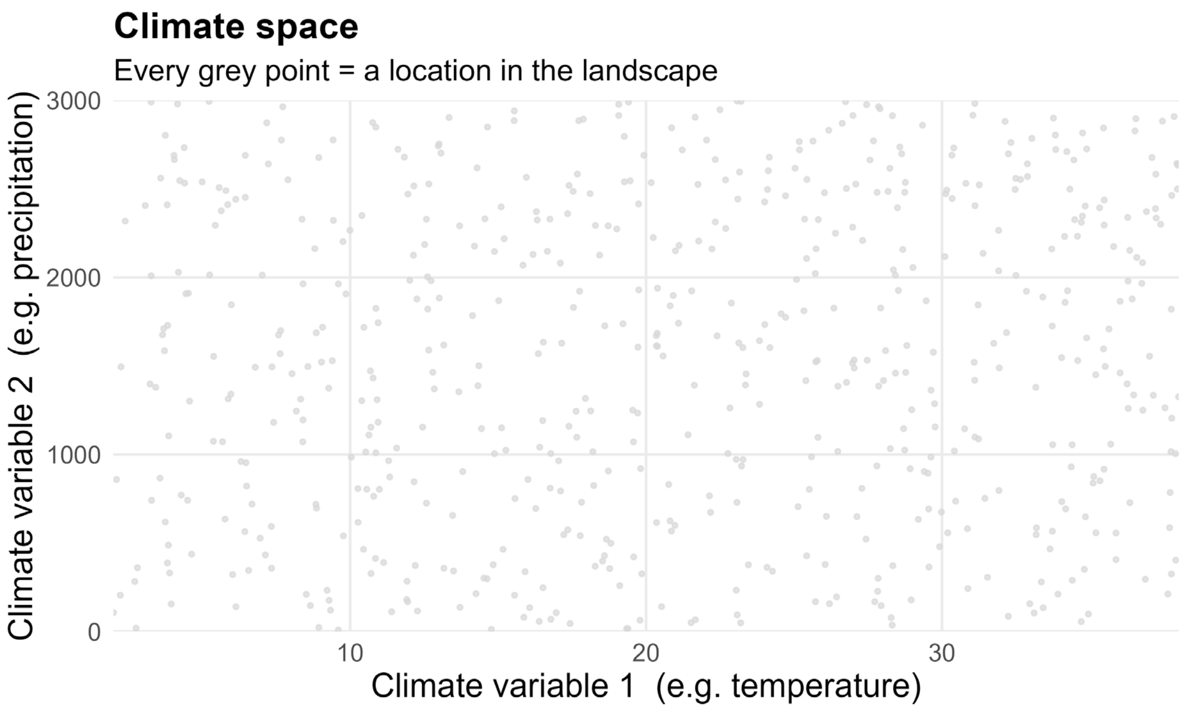




Limitations

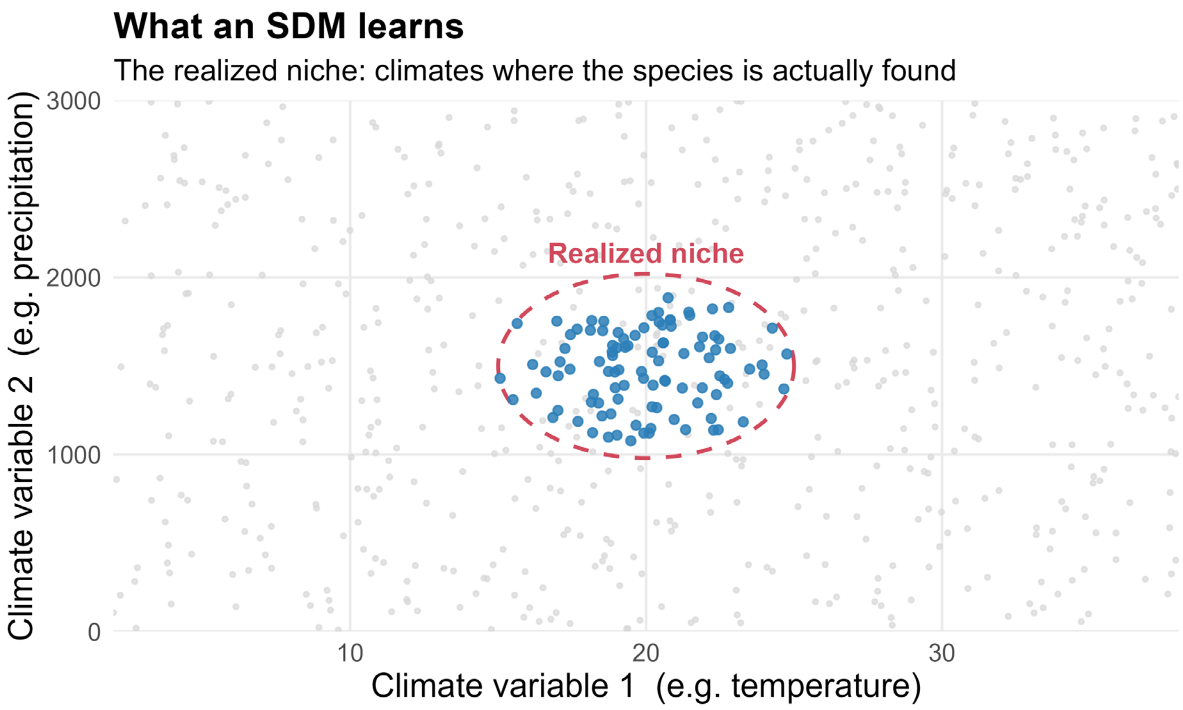
Biotic interactions, model bias, model assumptions

Limitations of environment-only models



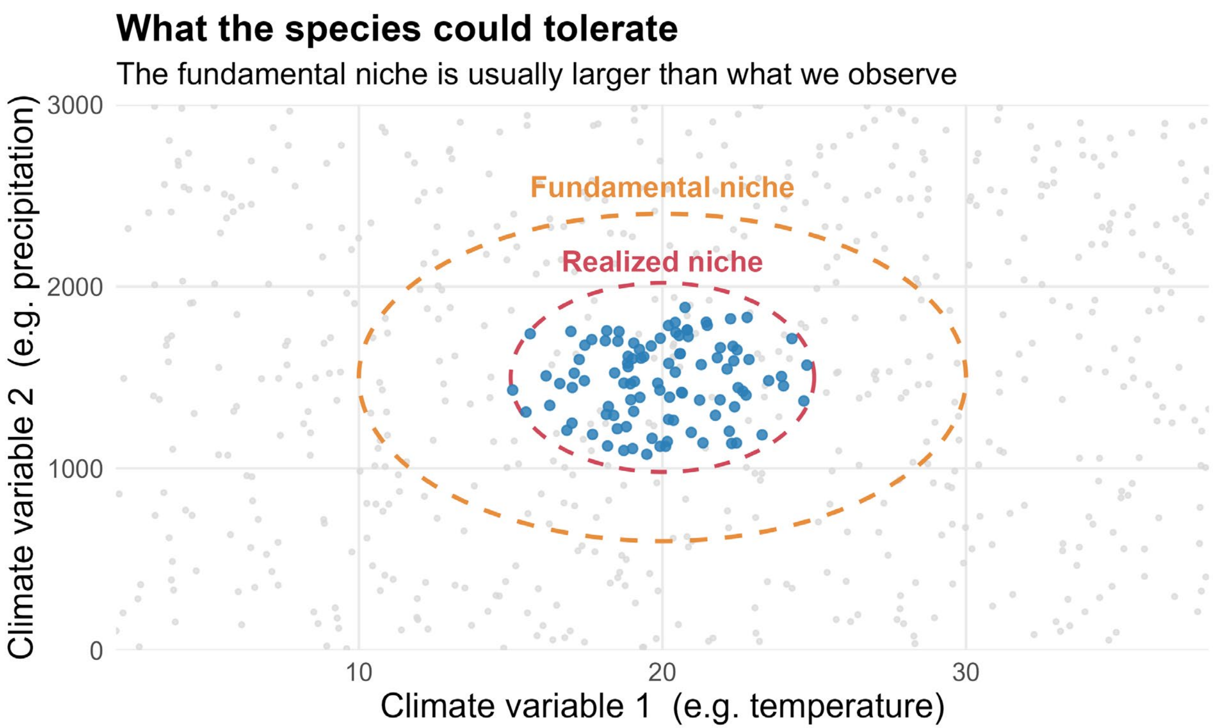
- SDMs aim to predict the **realised distribution** of a species — where it actually occurs
- To do this, we model the relationship between known occurrences and environmental variables
  - i.e. which environments contain suitable habitats

Limitations of environment-only models



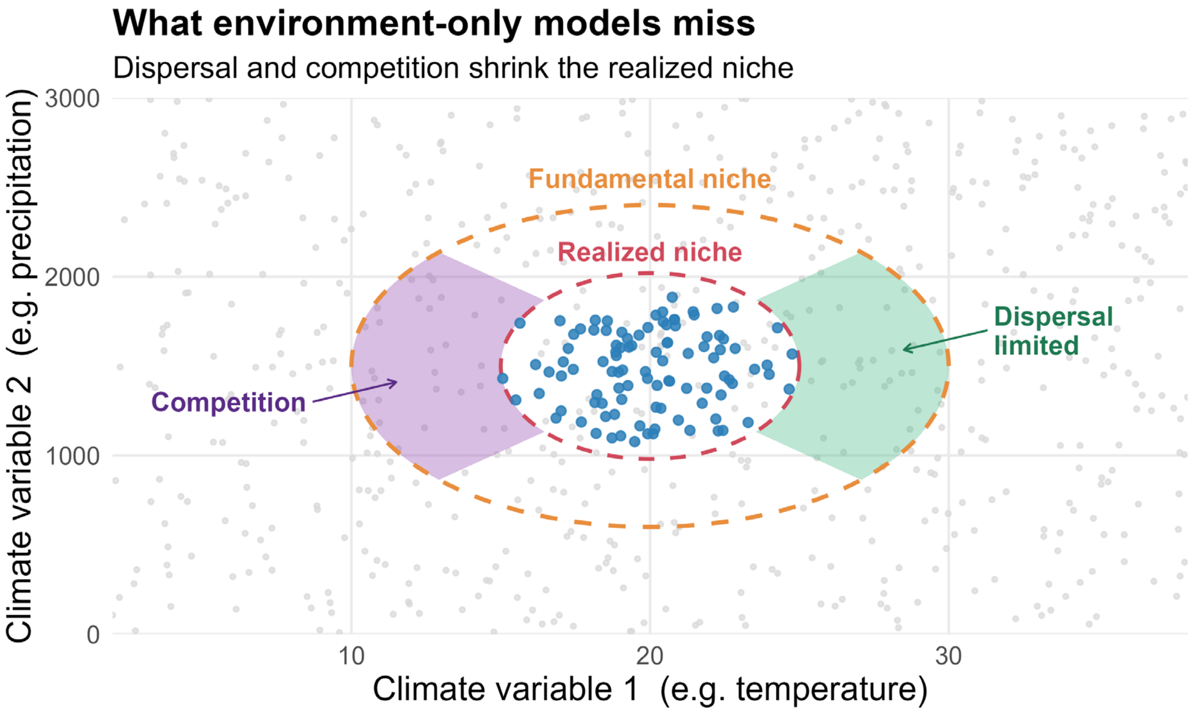
- But occurrence data only tells us about the **realised niche** — a subset of where a species *could* survive

Limitations of environment-only models



- The **fundamental niche**
  - the full set of environmental conditions under which a species can persist
- The **realised niche**:
  - the subset of those conditions where a species is actually found
- SDMs are trained on realised niche data, but the models implicitly try to estimate the fundamental niche

Limitations of environment-only models

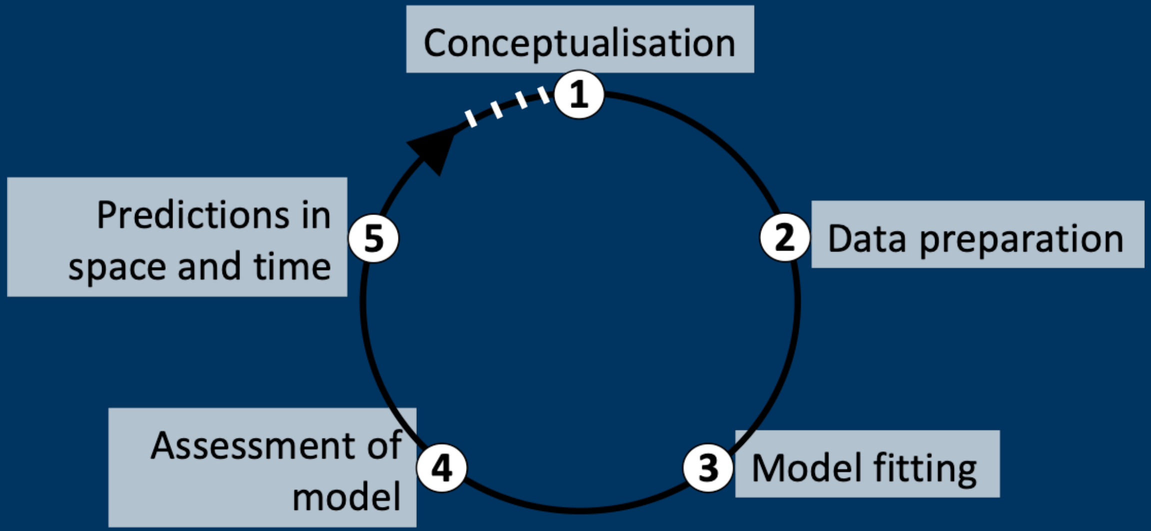


- **Dispersal barriers** — geographic features that prevent range expansion
- **Historical factors** — species may not have had time to colonise suitable areas
- **Biotic interactions** — competition, predation, or (absence of) mutualisms can exclude species from otherwise suitable habitat
- SDMs can struggle to distinguish *unsuitable* from *unreached*

Take Homes

- SDMs involve lots of decisions and it may be difficult to know how they will affect the outcome:
  - selection of background points
  - choice of predictor variables
  - choice of model algorithm
  - method of cross validation
  - choice of evaluation summary metrics
- Best practise is to explore alternatives and test the robustness of results
- This is true of all applications of data science and quantitative biology

SDM modelling cycle



Readings and Workshop Prep

(1) Valavi et al. 2021

Methods Paper

(2) Atlas of Living Australia

Online Lab

(3) Zurrell 2023

Online Lab

(4) Warren et al. 2020.

Methods paper

1. Please try to have all R packages installed before the workshop
2. Troubleshoot with Google/ChatGPT/Claude

References

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- *Shipley, B.R., Bach, R., Do, Y., Strathearn, H., McGuire, J.L. and Dilkina, B., 2022. megaSDM: integrating dispersal and time-step analyses into species distribution models. Ecography, 2022.*